

# YUCHEN DU

Phone: +1 (734) 596-5957 | Email: du313@purdue.edu | LinkedIn: <https://www.linkedin.com/in/yuchen-du/>

## SUMMARY

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**Ph.D. candidate** at Purdue University specializing in AI-driven optimization, transportation systems. Applied Scientist intern at Amazon developing multi-agent LLM systems; published in Transportation Research Part C&E (top transportation journals). Experienced in optimization (MIP/heuristic), AI/ML methods (RL, GANs, Transformers), system simulation, LLM agent.

## SKILLS

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- **Mathematical Optimization:** Mixed Integer Programming, Heuristics, Stochastic modeling.
- **AI/ML:** Reinforcement Learning, Generative Adversarial Networks, Transformers.
- **Project experience:** LLM agent system, Last mile delivery optimization. Transportation system simulation.
- **Programming:** Python (Gurobi, PyTorch, LangGraph), C# (Unity Godot), SQL

## EDUCATION

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| Purdue University, Ph.D at Industrial Engineering Technology                               | 2023-Expected July 2026 |
| University of Michigan, MSc in Data Science track at Industrial and Operations Engineering | 2021-2023               |
| South China University of Technology, BSc in Industrial Engineering                        | 2017-2021               |

## EXPERIENCE

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Amazon – Applied Scientist Intern Seattle, WA | Aug 2025-Nov 2025

- Developed a **three-stage LLM reasoning system (Learning → Graph → Processing)** for automating seller-appeal decisions using LangGraph.
- Built the **Learning System** to extract structured reasoning chains (Evidence → Action → Factor → Decision) from historical cases. Developed the **Graph System** for case-based learning, cross-case similarity merge, reasoning-pattern extraction across hundreds of historical cases.
- Implemented the **Processing System**, orchestrating a multi-agent pipeline for real-time case analysis: case understanding, similarity retrieval, factor generation, action verification, evidence extraction, and final decision recommendation.
- Demonstrated that the EAFD + graph-enhanced reasoning system enables LLM agents to **leverage historical case patterns to identify missing evidence and required investigative actions**, producing consistent, SOP-independent decisions and delivering clear, seller-facing explanations for each recommendation.

## PAPERS

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**“A two-stage stochastic optimization model for sidewalk robot food delivery systems”**

*Transportation Research Part E* (2025) [paper link](#)

- Developed a stochastic mixed integer programming + Adaptive Large Neighborhood Search optimization framework for multi-depot last-mile delivery; evaluated under traffic, battery, and capacity constraints to improve operational efficiency.

**“Analytical model for large-scale design of sidewalk delivery robot systems with bounded makespans”**

*Transportation Research Part C* (2024). [paper link](#)

- Developed a Continuous Approximation model for fleet sizing and cost optimization in large-scale sidewalk delivery robot (SDR) networks. Validated on 10K+ synthetic NYC data, identified significant cost savings for strategic SDR deployment.

**“Understanding Social Perception, Interactions, and Safety Aspects of Sidewalk Delivery Robots Using Sentiment Analysis”**, *Transportation Research Record* (2025). [paper link](#) | [github](#)

- Applied BERT, LSTM, and topic modeling to analyze 10K+ YouTube comments, identifying key concerns for public perception of SDRs. Provided data-driven policy insights, shaping recommendations for SDR regulations.

## PROJECTS

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**“Electric Vehicle Fleet Simulation Platform”**, 2025

- Built a city-scale EV routing and dispatch simulation with order generation, multi-vehicle coordination, and charging management using Python, OSMnx/NetworkX. Implemented real-time visualization and scalable scenario testing.

**“Pedestrian’s trajectory generation using Generative Adversarial Network”**, 2024.

- Developed a **Generative Adversarial Network (GAN)** enhanced with LSTM modules to generate **pedestrian trajectories**, using **max pooling** to capture global information. The model demonstrated better performance than traditional LSTM and RNN models, producing socially acceptable and realistic pedestrian paths.